

Customer Segmentation Analysis Using K-Means Clustering

1. Introduction

Customer segmentation is the process of dividing customers into groups based on similar characteristics and purchasing behavior. This helps businesses understand their customers and create targeted marketing strategies. In this project, K-Means Clustering was used to segment customers based on Annual Income and Spending Score.

2. Objective

The objective of this project is to identify distinct customer groups and analyze their spending behavior to support data-driven business decisions.

3. Dataset Description

The Mall Customer Segmentation dataset contains customer information including:

- Customer ID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1–100)

The dataset consists of 200 customer records.

4. Methodology

Data Preprocessing

- Loaded the dataset using Pandas.
- Checked for missing values.
- Removed the CustomerID column.
- Converted Gender into numerical values.

Exploratory Data Analysis

- Analyzed Age distribution.
- Analyzed Annual Income distribution.
- Analyzed Spending Score distribution.
- Visualized the relationship between Income and Spending Score.

Clustering

- Selected Annual Income and Spending Score as clustering features.

- Applied StandardScaler for feature normalization.
- Used the Elbow Method to determine the optimal number of clusters.
- Applied K-Means Clustering with five clusters.

5. Results

The K-Means algorithm successfully grouped customers into five distinct segments based on income and spending behavior. The clusters revealed different customer types such as premium customers, regular customers, potential customers, and budget-conscious customers.

6. Business Insights

- High-income, high-spending customers are the most valuable segment.
- High-income, low-spending customers can be targeted through personalized promotions.
- Low-income, high-spending customers respond well to marketing campaigns.
- Customer segmentation helps businesses improve marketing efficiency and customer engagement.

7. Conclusion

This project successfully applied K-Means Clustering to perform customer segmentation. The analysis identified meaningful customer groups that can help businesses design targeted marketing strategies, improve customer satisfaction, and increase revenue.